

NovaDiff — Baseline codebase

Baseline: NovaDiff-main

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI baseline codebase narrative

Repository overview

The NovaDiff-main repository is a TypeScript codebase with a focus on desktop application development using Vite and Tauri. The repository contains 39 files, with 23 resolved relative import edges and 10 cross-file call edges (heuristically determined). The top extensions are .ts, .tsx, .cjs, .json, .svg, .md, .css, .html, .jpg, .lock, .png, .rs, .toml, and the top first-level directories are src/, electron/, cli/, public/, scripts/, tests/, and the directory depth histogram shows that the deepest directory is at depth 3 with 12 files. The detected project stacks are node and package.json.

Architectural layout

The NovaDiff-main repository has a modular architecture with a clear separation of concerns between the frontend and backend. The frontend is built using Vite and Tauri, while the backend is written in Rust and uses the Cargo ecosystem for dependency management. The repository contains several subdirectories, including src/ for the frontend code, electron/ for the desktop application, cli/ for the command-line interface, public/ for static assets, scripts/ for build and run scripts, tests/ for test suites, and various other directories for different components and utilities.

Key subsystems

The key subsystems in the NovaDiff-main repository are the frontend, which is built using Vite and Tauri, and the backend, which is written in Rust and uses the Cargo ecosystem for dependency management. The frontend is responsible for rendering the user interface and handling user input, while the backend is responsible for managing the data and performing computations.

Dependency signals

The NovaDiff-main repository has a number of dependencies, including Vite, Tauri, and Cargo. These dependencies are managed through package managers such as npm and cargo, and they provide a number of benefits, including easy installation and updates, versioning, and compatibility with other packages.

Operational considerations

One operational consideration for the NovaDiff-main repository is the use of TypeScript for type checking and code analysis. This helps to catch errors early in the development process and ensures that the codebase is maintainable and scalable. Another operational consideration is the use of a command-line interface (CLI) for running the desktop application, which allows users to interact with the application from the terminal.

Documentation gaps

There are several documentation gaps in the NovaDiff-main repository, including a lack of documentation for the backend code and a need for more detailed documentation on how to run and build the application. Additionally, there is a need for more testing coverage to ensure that the application is robust and reliable.

Suggested onboarding and verification

For onboarding, it would be helpful to have more documentation on how to set up the development environment and get started with the codebase. Additionally, it would be useful to have more automated tests to ensure that the application is stable and performs as expected. For verification, it would be helpful to have more documentation on the architecture and design decisions made during the development process, as well as more thorough testing to ensure that the application meets its intended functionality.

Baseline codebase metrics

Baseline codebase metrics

Repository: NovaDiff-main

Path: /Users/iamgroot/Downloads/NovaDiff-main

Metric	Value
Indexed files	39
Symbol-bearing files	17
Relative import edges	23
Cross-file call edges (heuristic)	10

Top extensions

Extension Count

.ts	8
.tsx	8
.cjs	5
.json	4
.svg	3
(no ext)	2
.md	2
.css	1
.html	1
.jpg	1
.lock	1
.png	1
.rs	1
.toml	1

Top directories

Directory Files

src	15
.	13
electron	4
cli	3
public	2
scripts	1
tests	1

Detected project stacks

- **node:** package.json

Sample files

- .gitignore
- CONTRIBUTIONS.md
- LICENSE

- README.md
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/compare-runner.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/preload.cjs
- index.html
- nova-diff-desktop.jpg
- nova-diff-icon.png
- package-lock.json
- package.json
- public/tauri.svg
- public/vite.svg
- scripts/run-electron.cjs
- src/App.css
- src/App.tsx
- src/app/changedFilesTree.ts
- src/app/diffExcerpt.ts
- src/app/llmStorage.ts
- src/app/types.ts
- src/assets/react.svg
- src/components/ChangedFilesTree.tsx
- src/components/DiffWorkspace.tsx
- src/components/InsightsColumn.tsx
- src/components/LlmSettingsModal.tsx
- src/components/LlmSummaryMarkdown.tsx
- src/components/SidebarNav.tsx
- src/main.tsx
- src/vite-env.d.ts
- tests/smoke.test.ts
- tsconfig.json
- tsconfig.node.json
- vite.config.ts
- vitest.config.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, rel_path, hash_file, collect_files, compare_folders, read_utf8_file, line_diff_to_rows, get_file_diff, main, rel_path_normalizes
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine
electron/llm.cjs	normalizeBase, sendAccumulated, streamOllamaChat, streamLmStudioChat, buildSummarizePrompt, summarizeChangeStream, completeChat, summarizeChange, probeProvider
electron/main.cjs	iconPath, buildMenu, createWindow
scripts/run-electron.cjs	cleanEnv, resolveElectronBinary
src/App.tsx	isElectron, pickDirectory, baseName, clampInsightsWidth, loadInsightsWidth
src/app/changedFilesTree.ts	DirNode, FileNode, normalizePathSegments, ancestorDirPaths, sortChildren, ensureDir, addFilePathToTree, buildChangedFilesTreeRoot, flattenVisibleRows
src/app/diffExcerpt.ts	buildDiffExcerpt
src/app/llmStorage.ts	LlmSettings, LlmSummarizePayload, defaultBase, defaultModel, defaultLlmSettings, loadLlmSettings, saveLlmSettings
src/app/types.ts	FileChange, DiffRow, FileDiffPayload
src/components/ChangedFilesTree.tsx	ChangedFilesTreeProps, TreeRow, ChangedFilesTree
src/components/DiffWorkspace.tsx	DiffWorkspaceProps, DiffWorkspace, DiffTableRow
src/components/InsightsColumn.tsx	InsightsColumnProps, InsightsColumn, MetricPill
src/components/LlmSettingsModal.tsx	LlmSettingsModalProps, LlmSettingsModal
src/components/LlmSummaryMarkdown.tsx	LlmSummaryMarkdownProps, LlmSummaryMarkdown
src/components/SidebarNav.tsx	SidebarNavProps, SidebarNav
src/vite-env.d.ts	ElectronAPI

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
src/App.tsx	src/app/types.ts
src/App.tsx	src/app/diffExcerpt.ts
src/App.tsx	src/app/llmStorage.ts
src/App.tsx	src/components/DiffWorkspace.tsx
src/App.tsx	src/components/InsightsColumn.tsx
src/App.tsx	src/components/LlmSettingsModal.tsx
src/App.tsx	src/components/SidebarNav.tsx
src/app/changedFilesTree.ts	src/app/types.ts
src/app/diffExcerpt.ts	src/app/types.ts
src/components/ChangedFilesTree.tsx	src/app/types.ts
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts
src/components/DiffWorkspace.tsx	src/app/types.ts
src/components/InsightsColumn.tsx	src/app/types.ts
src/components/InsightsColumn.tsx	src/app/llmStorage.ts
src/components/InsightsColumn.tsx	src/components/LlmSummaryMarkdown.tsx
src/components/InsightsColumn.tsx	src/components/ChangedFilesTree.tsx
src/components/LlmSettingsModal.tsx	src/app/llmStorage.ts
src/components/SidebarNav.tsx	nova-diff-icon.png
src/main.tsx	src/App.tsx
src/vite-env.d.ts	src/app/types.ts
src/vite-env.d.ts	src/app/llmStorage.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
src/App.tsx	src/app/llmStorage.ts	loadLlmSettings
src/App.tsx	src/app/diffExcerpt.ts	buildDiffExcerpt
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	buildChangedFilesTreeRoot
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	flattenVisibleRows
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	ancestorDirPaths
src/components/LlmSettingsModal.tsx	src/app/llmStorage.ts	saveLlmSettings

Codebase outline (JSON excerpt)

```
{
  "by_extension": [
    {
      "count": 8,
      "ext": ".ts"
    },
    {
      "count": 8,
      "ext": ".tsx"
    },
    {
      "count": 5,
      "ext": ".cjs"
    },
    {
      "count": 4,
      "ext": ".json"
    },
    {
      "count": 3,
      "ext": ".svg"
    },
    {
      "count": 2,
      "ext": "(no ext)"
    },
    {
      "count": 2,
      "ext": ".md"
    },
    {
      "count": 1,
      "ext": ".css"
    },
    {
      "count": 1,
      "ext": ".html"
    },
    {
      "count": 1,
      "ext": ".jpg"
    },
    {
      "count": 1,
      "ext": ".lock"
    },
    {
      "count": 1,
      "ext": ".png"
    },
    {
      "count": 1,
      "ext": ".rs"
    },
    {
      "count": 1,
      "ext": ".toml"
    }
  ],
  "class_diagram_mermaid": "classDiagram\n  class ChangeKind\n  class ChangedFilesTree\n  class ChangedFilesTreeProps\n  class DiffRow\n  class DiffTableRow\n  class DiffWorkspace\n  class DiffWorkspaceProps\n  class DirNode\n  class ElectronAPI\n  class FileChange\n  class FileDiffPayload\n  class FileNode\n  class InsightsColumn\n  class InsightsColumnProps\n  class LlmSettings\n  class LlmSettingsModal\n  class LlmSettingsModalProps\n  class LlmSummarizePayload\n  class LlmSummaryMarkdown\n  class LlmSummaryMarkdownProps\n  class MetricPill\n  class SidebarNav\n  class SidebarNavProps\n  class TreeRow",
  "cross_file_call_edges": [
    {
      "from_file": "electron/main.cjs",
      "to_file": "electron/compare-runner.cjs",
      "via": "runCompareEngine"
    },
    {
      "from_file": "electron/main.cjs",
      "to_file": "electron/llm.cjs",
      "via": "summarizeChange"
    }
  ],
}
```

```

{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "summarizeChangeStream"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "probeProvider"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/llmStorage.ts",
  "via": "loadLlmSettings"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/diffExcerpt.ts",
  "via": "buildDiffExcerpt"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "buildChangedFilesTreeRoot"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "flattenVisibleRows"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "ancestorDirPaths"
},
{
  "from_file": "src/components/LlmSettingsModal.tsx",
  "to_file": "src/app/llmStorage.ts",
  "via": "saveLlmSettings"
}
],
"cross_file_call_graph_mermaid": "flowchart LR\n  electron_main_cjs[\"electron/main.cjs\"] -->|runCompareEngine| electron_compare_runner_cjs[\"electron/compare-runner.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|summarizeChange| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|probeProvider| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|summarizeChangeStream| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|loadLlmSettings| src_app_llmStorage_ts[\"src/app/llmStorage.ts\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|buildDiffExcerpt| src_app_diffExcerpt_ts[\"src/app/diffExcerpt.ts\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|buildChangedFilesTreeRoot| src_app_changedFilesTree_ts[\"src/app/changedFilesTree.ts\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|flattenVisibleRows| src_app_changedFilesTree_ts[\"src/app/changedFilesTree.ts\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|ancestorDirPaths| src_app_changedFilesTree_ts[\"src/app/changedFilesTree.ts\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|saveLlmSettings| src_app_llmStorage_ts[\"src/app/llmStorage.ts\"]",
  "detected_projects": [
    {
      "kind": "node",
      "markers": [
        "package.json"
      ]
    }
  ]
},
"files": [
  {
    "depth": 1,
    "ext": "(no ext)",
    "line_count": 0,
    "path": ".gitignore",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 1,
    "ext": ".md",
    "line_count": 0,
    "path": "CONTRIBUTIONS.md",
    "symbol_count": 0,
    "top_directory": "."
  }
],
{

```

```

    "depth": 1,
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    "path": "LICENSE",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
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    "ext": ".md",
    "line_count": 0,
    "path": "README.md",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 2,
    "ext": ".lock",
    "line_count": 0,
    "path": "cli/Cargo.lock",
    "symbol_count": 0,
    "top_directory": "cli"
  },
  {
    "depth": 2,
    "ext": ".toml",
    "line_count": 0,
    "path": "cli/Cargo.toml",
    "symbol_count": 0,
    "top_directory": "cli"
  },
  {
    "depth": 3,
    "ext": ".rs",
    "line_count": 332,
    "path": "cli/src/main.rs",
    "symbol_count": 10,
    "top_directory": "cli"
  },
  {
    "depth": 2,
    "ext": ".cjs",
    "line_count": 89,
    "path": "electron/compare-runner.cjs",
    "symbol_count": 4,
    "top_directory": "electron"
  },
  {
    "depth": 2,
    "ext": ".cjs",
    "line_count": 350,
    "path": "electron/llm.cjs",
    "symbol_count": 9,
    "top_directory": "electron"
  },
  {
    "depth": 2,
    "ext": ".cjs",
    "line_count": 192,
    "path": "electron/main.cjs",
    "symbol_count": 3,
    "top_directory": "electron"
  },
  {
    "depth": 2,
    "ext": ".cjs",
    "line_count": 50,
    "path": "electron/preload.cjs",
    "symbol_count": 0,
    "top_directory": "electron"
  },
  {
    "depth": 1,
    "ext": ".html",
    "line_count": 0,
    "path": "index.html",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 1,
    "ext": ".jpg",

```

```
    "line_count": 0,
    "path": "nova-diff-desktop.jpg",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 1,
    "ext": ".png",
    "line_count": 0,
    "path": "nova-diff-icon.png",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 1,
    "ext": ".json",
    "line_count": 0,
    "path": "package-lock.json",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 1,
    "ext": ".json",
    "line_count": 0,
    "path": "package.json",
    "symbol_count": 0,
    "top_directory": "."
  },
  {
    "depth": 2,
    "ext": ".svg",
    "line_count": 0,
    "path": "public/tauri.svg",
    "symbol_count": 0,
    "top_directory": "public"
  },
  {
    "depth": 2,
    "ext": ".svg",
    "line_count": 0,
    "path": "public/vite.svg",
    "symbol_count": 0,
    "top_directory": "public"
  },
  {
    "depth": 2,
    "ext": ".cjs",
    "line_count": 65,
    "path": "scripts/run-electron.cjs",
    "symbol_count": 2,
    "top_directory": "scripts"
  },
  {
    "depth": 2,
    "ext": ".css",
    "line_count": 0,
    "path": "src/App.css",
    "symbol_count": 0,
    "top_directory": "src"
  },
  {
    "depth": 2,
    "ext": ".tsx",
    "line_count": 378,
    "path": "src/App.tsx",
    "symbol_count": 5,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".ts",
    "line_count": 160,
    "path": "src/app/changedFilesTree.ts",
    "symbol_count": 9,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".ts",
    "line_count": 24,
    "path": "src/app/diffExcerpt.ts",
```

```

    "symbol_count": 1,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".ts",
    "line_count": 64,
    "path": "src/app/llmStorage.ts",
    "symbol_count": 7,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".ts",
    "line_count": 22,
    "path": "src/app/types.ts",
    "symbol_count": 3,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".svg",
    "line_count": 0,
    "path": "src/assets/react.svg",
    "symbol_count": 0,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 212,
    "path": "src/components/ChangedFilesTree.tsx",
    "symbol_count": 3,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 193,
    "path": "src/components/DiffWorkspace.tsx",
    "symbol_count": 3,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 182,
    "path": "src/components/InsightsColumn.tsx",
    "symbol_count": 3,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 158,
    "path": "src/components/LlmSettingsModal.tsx",
    "symbol_count": 2,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 33,
    "path": "src/components/LlmSummaryMarkdown.tsx",
    "symbol_count": 2,
    "top_directory": "src"
  },
  {
    "depth": 3,
    "ext": ".tsx",
    "line_count": 69,
    "path": "src/components/SidebarNav.tsx",
    "symbol_count": 2,
    "top_directory": "src"
  },
  {
    "depth": 2,
    "ext": ".tsx",
    "line_count": 9,
    "path": "src/main.tsx",
    "symbol_count": 0,
    "top_directory": "src"
  }

```

```

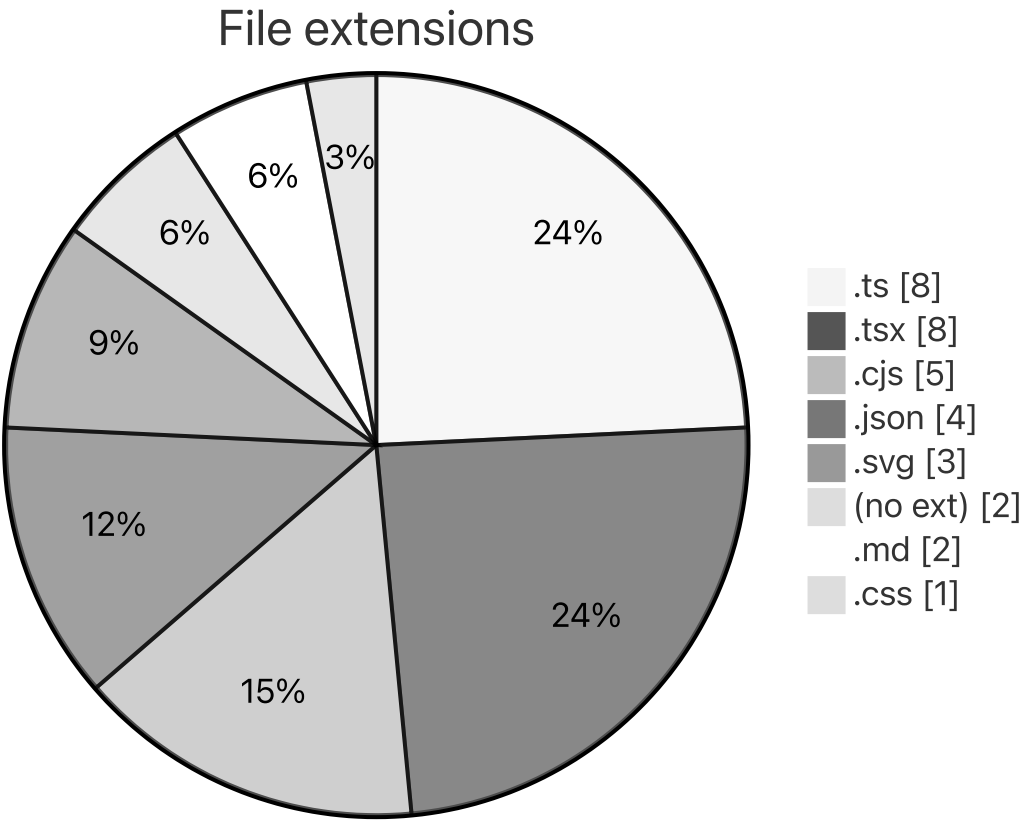
},
{
  "depth": 2,
  "ext": ".ts",
  "line_count": 31,
  "path": "src/vite-env.d.ts",
  "symbol_count": 1,
  "top_directory": "src"
},
{
  "depth": 2,
  "ext": ".ts",
  "line_count": 7,
  "path": "tests/smoke.test.ts",
  "symbol_count": 0,
  "top_directory": "tests"
},
{
  "depth": 1,
  "ext": ".json",
  "line_count": 0,
  "path": "tsconfig.json",
  "symbol_count": 0,
  "top_directory": "."
},
{
  "depth": 1,
  "ext": ".json",
  "line_count": 0,
  "path": "tsconfig.node.json",
  "symbol_count": 0,
  "top_directory": "."
},
{
  "depth": 1,
  "ext": ".ts",
  "line_count": 17,
  "path": "vite.config.ts",
  "symbol_count": 0,
  "top_directory": "."
},
{
  "depth": 1,
  "ext": ".ts",
  "line_count": 8,
  "path": "vitest.config.ts",
  "symbol_count": 0,
  "top_directory": "."
}
],
"import_edges": [
  {
    "from": "electron/main.cjs",
    "to": "electron/compare-runner.cjs"
  },
  {
    "from": "electron/main.cjs",
    "to": "electron/llm.cjs"
  },
  {
    "from": "src/App.tsx",
    "to": "src/app/types.ts"
  },
  {
    "from": "src/App.tsx",
    "to": "src/app/diffExcerpt.ts"
  },
  {
    "from": "src/App.tsx",
    "to": "src/app/llmStorage.ts"
  },
  {
    "from": "src/App.tsx",
    "to": "src/components/DiffWorkspace.tsx"
  },
  {
    "from": "src/App.tsx",
    "to": "src/components/InsightsColumn.tsx"
  },
  {
    "from": "src/App.tsx",
    "to": "src/components/LlmSettingsModal.tsx"
  }
]

```

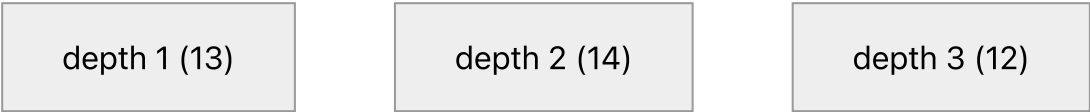
```
  },
  {
    "from": "src/App.tsx",
    "to": "src/components/SidebarNav.tsx"
  },
  {
    "from": "src/app/changedFilesTree.ts",
    "to": "src/app/types.ts"
  },
  {
    "from": "src/app/diffExcerpt.ts",
    "to": "src/app/types.ts"
  },
  {
    "from": "src/components/ChangedFilesTree.tsx",
    "to": "src/app/types.ts"
  },
  {
    "from": "src/components/ChangedFilesTree.tsx",
    "to": "src/app/changedFilesTree.ts"
  },
  {
    "from": "src/components/DiffWorkspace.tsx",
    "to": "src/app/types.ts"
  },
  {
    "from":
... truncated ...
```

Diagrams (Mermaid)

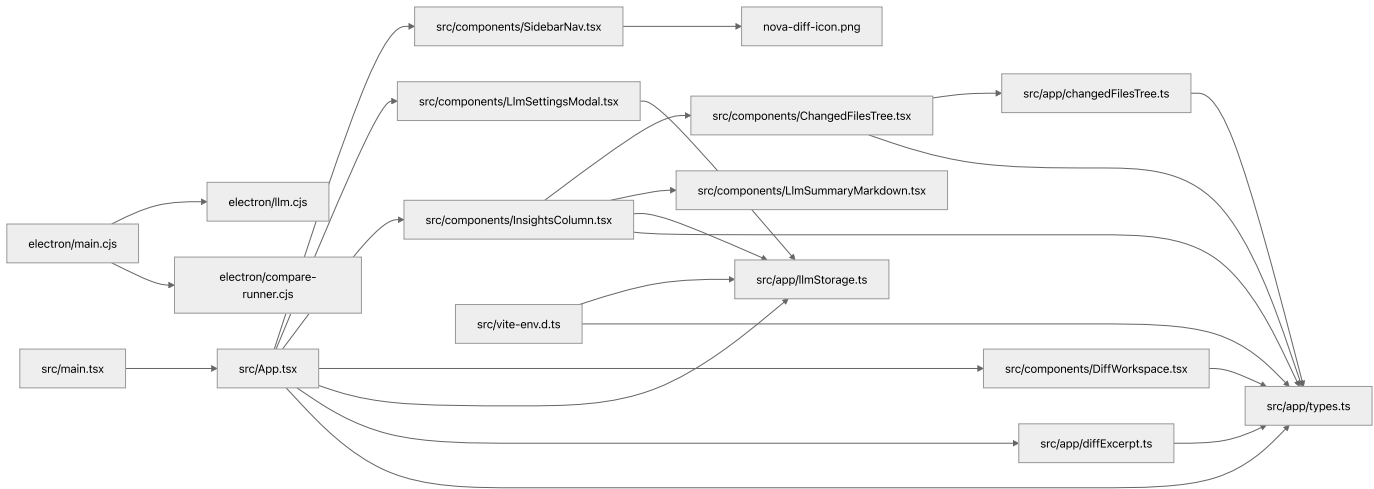
File extension mix



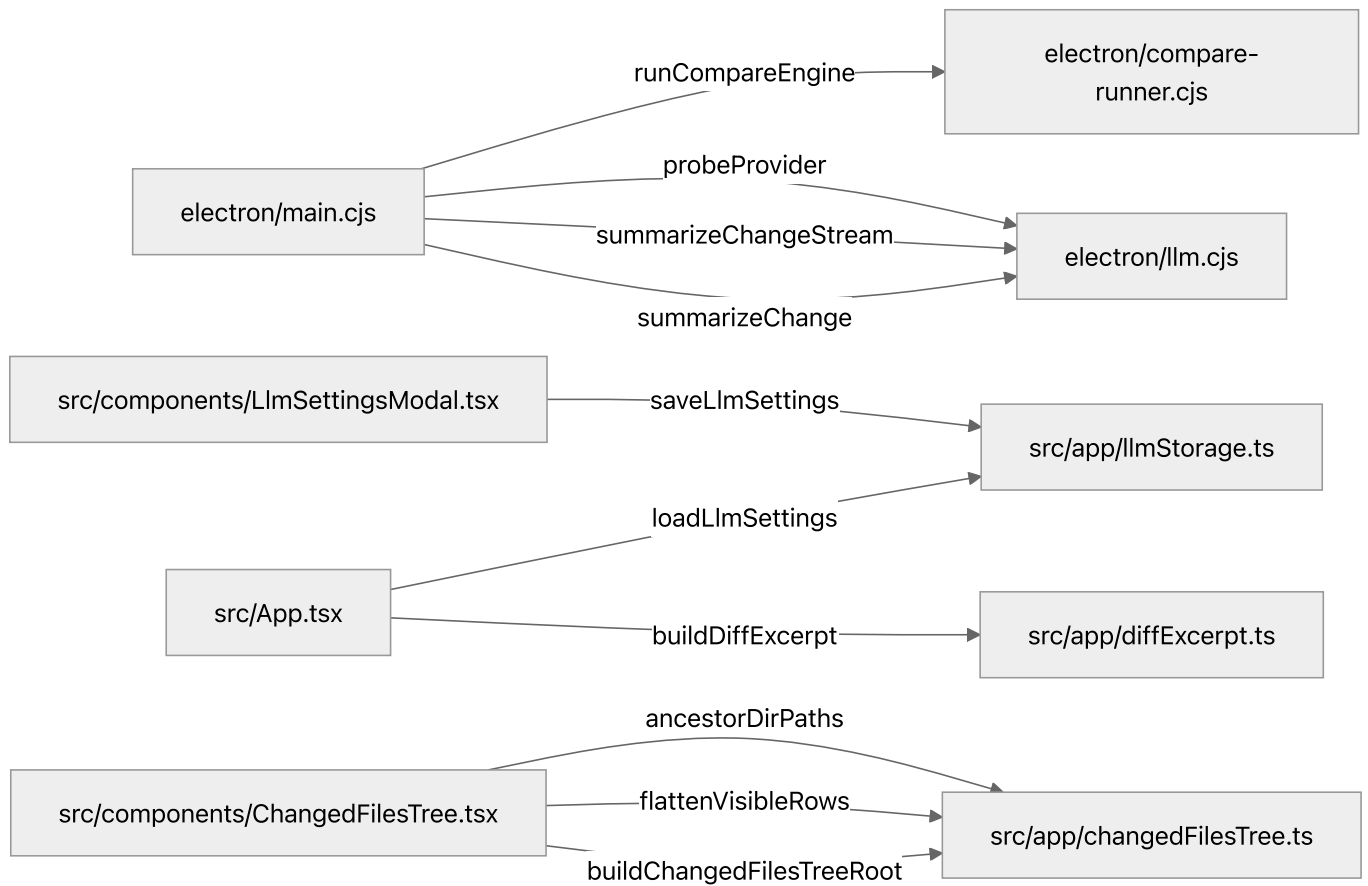
Directory depth



Module import graph (relative imports)



Cross-file calls (heuristic)



Types / classes (regex sketch)

